

Giant Pizza

Time limit: 1.00 s

Memory limit: 512 MB

Uolevi's family is going to order a large pizza and eat it together. A total of n family members will join the order, and there are m possible toppings. The pizza may have any number of toppings.

Each family member gives two wishes concerning the toppings of the pizza. The wishes are of the form "topping x is good/bad". Your task is to choose the toppings so that at least one wish from everybody becomes true (a good topping is included in the pizza or a bad topping is not included).

Input

The first input line has two integers n and m : the number of family members and toppings. The toppings are numbered $1, 2, \dots, m$.

After this, there are n lines describing the wishes. Each line has two wishes of the form "+ x " (topping x is good) or "- x " (topping x is bad).

Output

Print a line with m symbols: for each topping "+" if it is included and "-" if it is not included. You can print any valid solution.

If there are no valid solutions, print "IMPOSSIBLE".

Constraints

- $1 \leq n, m \leq 10^5$
- $1 \leq x \leq m$

Example

Input	Output
3 5 + 1 + 2 - 1 + 3 + 4 - 2	- + + + -